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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) Regular EXAMINATION October 2026

FIFTH SEMESTER

Discipline Specific Core Course (DSC) - MG5SECGEO300 -GEOINFORMATICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Answer in 60 words

Answer any **7** questions

Each question carries **2** marks

Q1. Principal Point in Aerial Photograph: The point where the camera's optical axis intersects the photograph, can be identified with the help of fiducial marks (1 mark). It represents the geometric centre of the photograph (1 mark). [U] / [CO1]

Q2. Effects of Rayleigh Scattering. [U] / [CO1]

Shorter wavelengths scatter more strongly (1 mark), making the sky appear blue. It causes haze in the remote sensing data products (1 mark).

Q3. IRS satellite. [U] / [CO1]

Indian Remote Sensing satellite series-mention 2-5 satellites (1 mark), used for resource mapping, agriculture, forestry and environmental monitoring (1 mark).

Q4. Zone Delineation. [K] / [CO4]

Zone delineation in GIS is the process of defining boundaries that divide geographic space into meaningful areas or zones. A zone is a spatial area enclosed by a boundary in which features tend to have similar characteristics (1 mark). For example, Hydrological Zones is an example, which include Watersheds, Sub-watersheds, Drainage basins, Riparian zones, Flood-prone areas etc. (1 mark).

Q5. ERDAS Imagine. [U] / [CO3]

ERDAS (Earth Resource Data Analysis) - ERDAS is a commercial GIS software used primarily as an image processing software, that has been expanded to include significant GIS and 3D capabilities (1.5 marks). It consists of various modules such as Imagine Advantage, Imagine Professional, and Virtual GIS (0.5 Marks).

Q6. Vector Data. [U] / [CO2]

Spatial data represented by points, lines and polygons (1 mark), used to depict features such as locations, roads and boundaries (1 mark).

Q7. Role of GCP in georeferencing. [A] / [CO3]

Ground Control Points (GCPs) are identifiable locations with known ground coordinates (1 mark). They are used to align satellite imagery or aerial photographs with a map coordinate system, ensuring accurate spatial positioning and reducing geometric distortions (1 mark).

Q 8. Quantitative mapping

Quantitative mapping is the graphical representation of numerical geographical data, such as population, rainfall, temperature and mineral distribution, using proportional symbols, choropleth maps, dot maps and isolines (Marks 2).

Part B: Short Notes

Each question carries 8 marks | Answer any 3

Q10. Mirror Stereoscope and Pocket Stereoscope (8 Marks)

- Definition – 1 mark: Instruments used to view overlapping aerial photographs in 3D.
- Mirror stereoscope – 2 marks: Uses mirrors and prisms; provides a wider viewing area and supports detailed interpretation.
- Pocket stereoscope – 2 marks: Uses two lenses; portable, inexpensive and suitable for quick observations.
- Comparison – 2 marks: Mirror type is larger and suited to detailed work; pocket type is compact with a limited viewing area.
- Conclusion – 1 mark: Both are useful in photogrammetry and terrain interpretation.

Q11. Map Projection and Types (8 Marks)

- Definition – 2 marks: Mathematical transformation of the curved Earth onto a flat map, inevitably introducing distortions.
- Cylindrical projection – 1.5 marks: Uses a cylinder; useful for world maps and navigation.
- Conical projection – 1.5 marks: Uses a cone; suitable for middle-latitude regions.
- Azimuthal projection – 1.5 marks: Uses a plane; useful for polar regions and directional mapping.
- Properties and applications – 1.5 marks: Conformal preserves angles, equal-area preserves area, and equidistant preserves selected distances.

Q12. Overlay Analysis in GIS (8 Marks)

- Definition – 2 marks: Combines multiple spatial layers to identify relationships and patterns.
- Types – 2 marks: Vector overlay, raster overlay and weighted overlay.
- Procedure – 1 mark: Prepare datasets, align coordinate systems, select layers and perform overlay.
- Applications – 2 marks: Site selection, groundwater potential, landslide zonation and land-use planning.
- Limitations and conclusion – 1 mark: Accuracy depends on input data quality, resolution and appropriate weighting.

Part C: Essay

Each question carries 12 marks | Answer any 1

Q14. GPS: Principles, Components and Applications (12 Marks)

- Introduction – 1 mark: GPS is a satellite-based positioning, navigation and timing system.
- Basic principles – 3 marks: Satellite signals, travel-time measurement, trilateration and use of at least four satellites for standard 3D positioning with clock correction.
- Major components – 3 marks:
    - Space segment – 1 mark
    - Control segment – 1 mark
    - User segment – 1 mark

- Applications – 4 marks: Surveying, GIS mapping, geological mapping, remote sensing, disaster management, precision agriculture and crustal deformation monitoring.
- Conclusion and limitations – 1 mark: GPS improves spatial data collection; atmospheric errors, multipath and signal obstruction can reduce accuracy.

Q15. Applications of GIS in Urban Planning and Management (12 Marks)

- Introduction – 1 mark: GIS collects, manages, analyses and displays spatial data for urban planning.
- Land-use planning – 2 marks: Zoning, urban expansion monitoring